

Standardization of NaOH Titrant and Volumetric Determination of Unknown Acid

Duy Hoang

Experiment Date: November 24, 2025

Report Due Date: December 2, 2025

Abstract

In this experiment, sodium hydroxide solution was standardized using potassium hydrogen phthalate as a primary standard and then followed by determination of an unknown concentration of a diprotic acid by pH-metric titration. In Part 1, a standard of about 0.10 M NaOH solution was prepared and calibrated against KHP with the help of phenolphthalein indicator. Three titrations with blank correction gave a mean NaOH concentration of (0.1066 ± 0.0009) M and a relative standard deviation of 0.35%. Part 2 involved titration of 25.00 mL aliquots of unknown diprotic acid with standardized NaOH and pH was recorded continuously. Initial titration established two equivalence points around 19 mL and exhibited typical diprotic acid behavior. The correct titration using smaller steps towards equivalence points yielded a full titration curve. The curve was analyzed and the first point of equivalence was found at 12.64 mL (pH 7.82) and the second at about 13.0 mL (pH 10.52). The first equivalence point volume was used to determine the unknown acid concentration. The pKa values were estimated by the use of the buffer regions between the equivalence points whereby $\text{pK}_{\text{a}1}$ was 2.6 and $\text{pK}_{\text{a}2}$ 10.5. Both sections displayed excellent accuracy and correct volumetric methodology necessary in the quantitative analysis of acids and bases.

Experimental

Reagents and Equipment

- Primary standard grade potassium hydrogen phthalate (KHP, MM = 204.221 g/mol)
- 10% sodium hydroxide stock solution, deionized water
- Phenolphthalein indicator solution, unknown diprotic acid sample

- 50 ml burette (± 0.05 ml)
- 250.0 ml volumetric flask (± 0.12 ml)
- 500.0 ml volumetric flask (± 0.2 ml)
- 25.00 ml volumetric pipette (± 0.03 ml)
- 250 ml Erlenmeyer flasks
- Ph meter with combination electrode
- Ph buffer solutions (ph 4, 7, and 10)
- Magnetic stirrer with stir bars
- Analytical balance (± 0.0001 g)
- Parafilm

Part 1: Standardization of NaOH Solution

KHP Standard Solution Preparation

The mass of KHP required for 250.00 mL of about 0.100 M solution was determined by the equation: $\text{mass} = (0.100 \text{ mol/L}) (0.250 \text{ L}) (204.221 \text{ g/mol}) = 5.1055 \text{ g}$. The KHP was qualitatively and quantitatively measured on an analytical balance and transferred to a 250.0 ml volumetric flask. The solid was dissolved in about 15 mL of deionized water and the beaker was

washed thrice with 15 mL portions of deionized water. Rinses were added to the volumetric flask. Deionized water was used to dilute the solution up to the calibration mark and mix it thoroughly by inverting 20 times. The precise concentration was determined using the actual mass.

NaOH Solution Preparation

The stock volume of 10% NaOH required for 500.00 mL of 0.10 M NaOH was determined. Density based molarity of 10% NaOH is about 2.5 M, and is given by: $(10 \text{ g}/100 \text{ g solution})/(40.0 \text{ g/mol})/\text{solution volume}$. To calculate the required volume, the dilution equation $M_1 V_1 = M_2 V_2$ was used: $V_1 = (0.100 \text{ M})(500.0 \text{ mL})/(2.5 \text{ M}) = 20 \text{ mL}$. Precisely, 20.0 mL of 10% NaOH stock was measured using a volumetric pipette and diluted and mixed with deionized water up to the mark in a 500.0 mL volumetric flask. The solution was properly mixed and placed in a closed vessel to reduce absorption of CO_2 in the atmosphere.

Titration Procedure

The 50 mL burette was washed three times with small amounts of the prepared NaOH solution and filled without any air bubbles in the tip. Parafilm covered the burette to avoid the entry of CO_2 . A single titration was done by measuring 25.00 mL of standard KHP solution in a 250 mL Erlenmeyer flask, adding 3 drops of phenolphthalein indicator and after which quick titration was done to the pink endpoint. This provided a rough final volume of 24.55 mL.

Three correct titrations were done by following the same procedure with careful drop-wise addition at the endpoint. The titration was terminated when the initial faint pink color lasted 20 seconds. At the endpoint, half-drops were added by creating a partial drop on the tip of the burette and transferring it to the wall of the flask. The sides of the flask were washed with deionized water. First and last readings of burette were noted to $\pm 0.01 \text{ mL}$. Blank titration was

conducted utilizing 25.00 mL deionized water with indicator, and 0.76 mL was obtained, and then deducted from each accurate titration volume.

Part 2: Unknown Diprotic Acid Titration

pH Electrode Calibration

The pH electrode was rinsed with deionized water and calibrated using two-point calibration. The electrode was inserted in a pH 7.00 buffer (green-yellow), meter adjusted to the calibration mode and the electrode allowed to stabilize at 7.00. The electrode was rinsed, put into pH 4.01 buffer (pink) and calibration was done when the reading stabilized. The electrode was kept in deionized water in between the measurements.

Preliminary Titration

A clean burette was filled with the standardized solution of NaOH. The 25.00 mL of the unknown diprotic acid was transferred into the 250 mL Erlenmeyer flask which had a magnetic stir bar. The flask was set on a magnetic stirrer with the pH electrode to ensure that the bulb was in the solution and never touched the flask walls or stir bar. The starting PH and burette reading were noted. Approximately 1.00 mL steps of NaOH were added to the solution and the pH was measured at each step. Titration was carried out until pH went beyond 11.5. The initial run identified two points of equivalence and informed the appropriate titration plan.

Accurate Titration

The unknown acid was titrated with the standardized NaOH solution in a fresh 25.00 mL aliquot of unknown acid. To start with, increments of 1.00 mL were added until the first part of the equivalence point was in the range of 2 mL. The increment at the buffer and equivalent point (2.5-5.5 pH-10-11) were lowered to 0.20 mL. Past the first equivalence point, the increments were taken back to larger volumes up to the point of approaching the second equivalence point

where 0.20 mL increments were once again employed. At every addition, the pH was recorded upon stabilization. The titration proceeded until the pH was over 11.9.

Results and Discussion

Part 1: NaOH Standardization

Titration Data and Calculations

The actual mass of the used solution was used to calculate the exact concentration of the KHP standard solution. The stoichiometric relationship between the molarity of the NaOH and the blank-corrected volume of NaOH was calculated:

$$M_{NaOH} = \frac{M_{KHP} V_{KHP}}{V_{NaOH \text{ corrected}}}$$

where the reaction stoichiometry is 1:1 for $\text{NaOH} + \text{KHP} \rightarrow \text{NaKP} + \text{H}_2\text{O}$.

Table 1: NaOH Standardization Titration Results

Run	Initial Burette (mL)	Final Burette (mL)	Volume NaOH (mL)	Corrected Volume (mL)	[NaOH] (M)
Crude	1.95	26.50	24.55	23.79	0.1051
T1	1.05	25.31	24.26	23.50	0.1064
T2	0.19	24.30	24.11	23.35	0.1071
T3	10.21	34.45	24.24	23.48	0.1065

Blank	6.46	7.22	0.76	0.76	3.289
-------	------	------	------	------	-------

Sample calculation [NaOH] (Trial 1)

Volume of NaOH=Final-Initial reading=25.31-1.05=24.26mL

Corrected volume = *Volume NaOH - Blank volume*

=24.26-0.76=23.50mL

$$M_{NaOH} = \frac{M_{KHP} V_{KHP}}{V_{NaOH \text{ corrected}}}$$

$M_{KHP}=0.100M$

$V_{KHP}=25.00mL$

$V_{NaOH \text{ corrected}}=23.50mL$

$$M_{NaOH} = \frac{0.100M \times 25.00mL}{23.50mL} = 0.1064M$$

Statistical Analysis

The mean NaOH concentration was calculated as:

$$Mean = \frac{Trial\ 1 + Trial\ 2 + Trial\ 3}{3}$$

$$Mean = \frac{0.1064 + 0.1071 + 0.1065}{3} = 0.1066M$$

The standard deviation was calculated using:

$$S = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$$

$$= \sqrt{\frac{(0.1064-0.1066)^2 + (0.1071-0.1066)^2 + (0.1065-0.1066)^2}{3-1}} = 0.000371M$$

The relative standard deviation:

$$RSD = \frac{s}{\bar{x}} \times 100 = \frac{0.000371}{0.1066} \times 100 = 0.35\%$$

The 95% confidence interval was calculated using the t-value for $n-1 = 2$ degrees of freedom ($t_{0.05, 2} = 4.303$):

$$\begin{aligned} CI &= \bar{x} \pm \frac{t.s}{\sqrt{n}} \\ &= 0.1066 \pm \frac{4.303 \times 0.000381}{\sqrt{3}} \\ &= 0.1066 \pm 0.0009M \end{aligned}$$

Therefore, $[\text{NaOH}] = (0.1067 \pm 0.0009) \text{ M}$

The great accuracy ($RSD < 1$ percent) proves the good volumetric technique and the careful determination of the endpoint. The three titrations concurred within 0.6, which indicated consistency in the execution of the experiment. The blank correction of 0.76 mL represents the little volume of NaOH needed to effect the color change of the phenolphthalein in pure water where dissolved CO_2 is present.

Part 2: Unknown Diprotic Acid Determination

Titration Curve Analysis

Initial titration results indicated the typical two-equivalent point diprotic acid titration curve. The pH was initially 1.91, meaning that there was a high concentration of acid or acid component. The first significant pH shift was between 18.98 mL and 20.02 mL (pH 4.59 to 10.17) and the second minor shift was between 20.02 mL and 21.01 mL (pH 10.17 to 11.30).

Table 2: Preliminary run

Vi(mL)	pH
--------	----

7.95	1.91
8.95	1.99
10.00	2.07
10.97	2.15
12.00	2.27
12.99	2.45
14.00	2.80
15.03	3.24
16.03	3.57
17.01	3.84
18.05	4.17
18.98	4.59
20.02	10.17
21.01	11.30
Vi(mL)	pH
0.95	2.23
1.90	2.21
3.03	2.22
4.00	2.24
4.98	2.35
5.51	2.39
5.75	2.43

Optimized titration using smaller steps gave a superior resolution of both the equivalence points and buffer zones. Figure 1 shows the titration curve. The first equivalence point was observed to be well defined at $V_1 = 12.64$ mL where the $pH = 7.82$. At this stage, the maximum slope (dpH/dV) was obtained, which indicated the full conversion of H_2A to HA^- . The second equivalence point was at $V_2 = 13.0$ mL $pH = 10.52$, representing total conversion of HA^- to A^{2-} .

Table 3: Accurate Titration

5.99	2.48
6.20	2.54
6.39	2.62
6.61	2.69
6.80	2.77
7.00	2.85
7.20	2.94
7.41	3.04
7.76	3.19
8.71	3.49
9.75	3.79
10.75	4.09
10.98	4.17
11.19	4.26
11.49	4.38
11.70	4.51
11.90	4.63
12.10	4.80
12.40	5.30
12.64	7.82
12.95	10.52
13.19	10.86

13.41	11.08
13.60	11.19
13.80	11.28
14.80	11.55
15.78	11.70
16.75	11.80
18.09	11.86
19.01	11.91

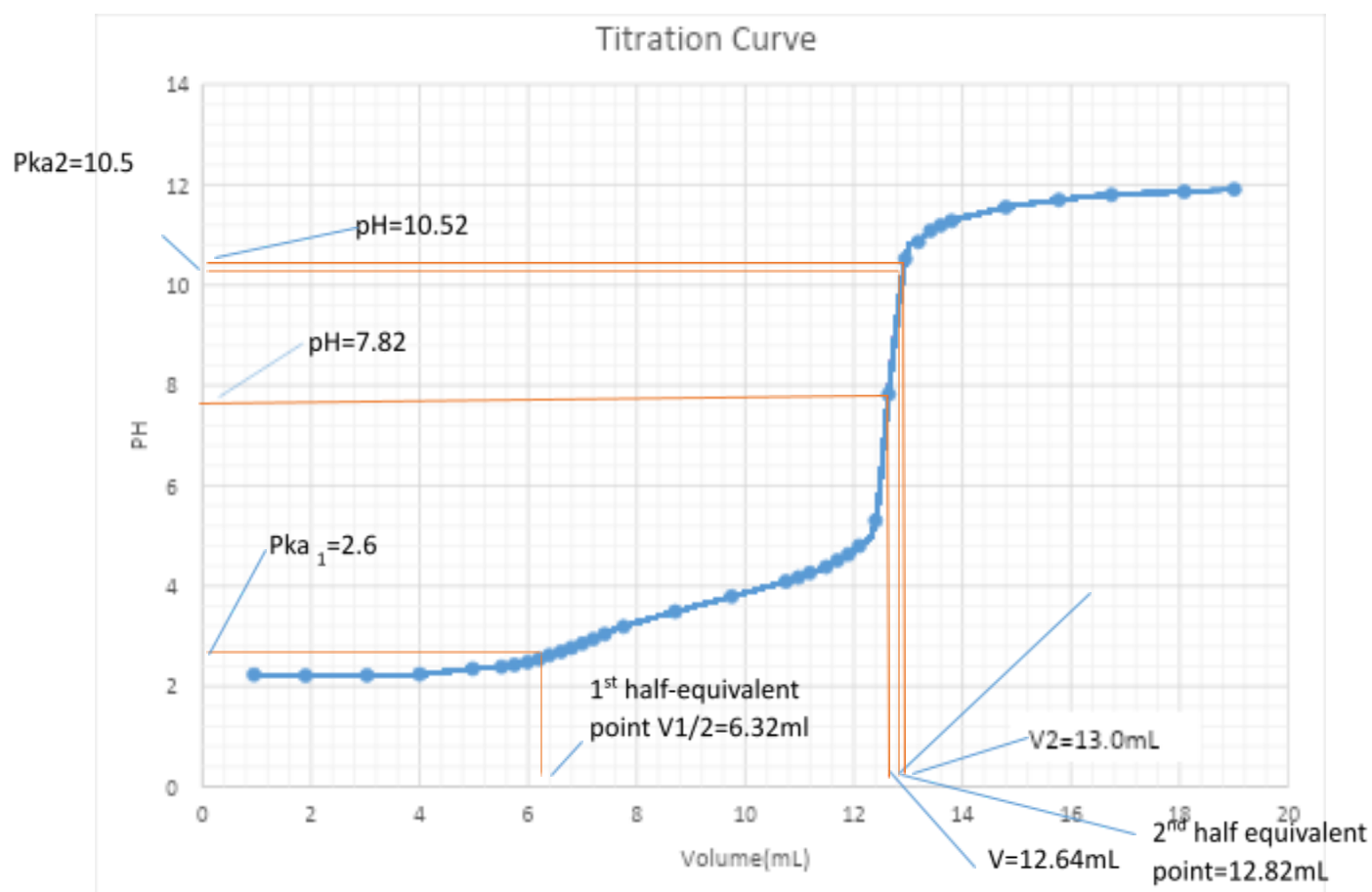


Figure 1: Titration Curve of Unknown Diprotic Acid with Standardized NaOH

Unknown Acid Concentration Calculation

The first equivalent point was used to determine the concentration of the unknown diprotic acid by the equation:

$$M_{acid} = \frac{M_{NaOH} \times V_{eq1}}{V_{acid}}$$

where:

- $M_{NaOH} = 0.1009 \text{ M}$
- $V_{eq1} = 12.64 \text{ mL}$ (first equivalence point)
- $V_{acid} = 25.00 \text{ mL}$

$$M_{acid} = \frac{0.1009M \times 12.64mL}{25.00mL} = 0.0510M$$

Unknown Acid Concentration: 0.0510M

pKa Estimation

The pKas were determined using the buffer sections of the titration curve. In the case of a diprotic acid, pKa₁ is the pH at the midpoint between the start and the first equivalent point (half-equivalence point), and pKa₂ is the pH at the midpoint between the first equivalent point and the second equivalent point.

First half-equivalence point: $V = V_1/2 = 12.64/2 = 6.32 \text{ mL}$

From the data, at approximately 6.32 mL, pH = 2.62, thus pKa₁ ≈ 2.6

Second half-equivalence point: $V = V_1 + (V_2 - V_1)/2 = 12.64 + (13.0 - 12.64)/2 = 12.82 \text{ mL}$

From the data, at approximately 12.95 mL, pH = 10.52, thus pKa₂ ≈ 10.5

These pKa values indicate that the unknown acid is probably oxalic acid ($\text{H}_2\text{C}_2\text{O}_4$) with literature pKa of 1.23 and 4.19 or a related dicarboxylic acid. Nonetheless, the experimental pKa2 is greater than that of oxalic acid, which could be indicative of malonic acid or other diprotic acid whose pKa values are widely separated.

Data Quality Assessment

The correct titration gave 37 data points with a superb coverage of the buffer regions and equivalence points. Regular inflection points and smooth curve show good calibration and equilibration time between additions of electrodes. The initial equivalence point was characterized by the sharp decrease in pH between 5.30 and 7.82 due to the addition of 0.24 mL, indicating that the buffering capacity was passed and the equivalent point was clearly identified.

The second equivalence point was less pronounced, increasing between 7.82 and 10.52 with an increase of 0.31 mL, characteristic of weak acid weak base systems where the HA^- species is a weak acid. The last pH value of 11.91 at 19.01 mL represents an excess of strong base after the second equivalence point.

Discussion Questions

1. Why is KHP suitable as a primary standard?

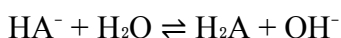
KHP is a good primary standard since it suits all the necessary requirements. It is highly pure (>99.9), does not absorb atmospheric moisture, and is not affected by drying in high temperatures, it has a high molar mass (204.221 g/mol) which reduces weighing errors, and is non-hygroscopic. These properties guarantee that the correct measurement of mass directly results in the correct molar quantities without further standardization.

2. Why must atmospheric CO_2 be minimized in this experiment?

Basic NaOH solutions dissolve CO_2 to give sodium carbonate (Na_2CO_3) or sodium bicarbonate (NaHCO_3) depending on the reaction: $\text{CO}_2 + 2\text{NaOH} \rightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O}$ or $\text{CO}_2 + \text{NaOH} \rightarrow \text{NaHCO}_3$. This reacts with NaOH and alters its concentration as time goes by. Furthermore, carbonic acid (H_2CO_3) dissolved in water can interfere with the end point, resulting in a loss of pink colour of the phenolphthalein during titration. These interferences can be minimized by using boiled deionized water, preparing NaOH with carbonate-free 10% stock, and storing solutions in a closed system.

3. What is the significance of the pH at the equivalence points?

The initial equivalence point pH of 7.82 is alkaline since the HA^- species is a stronger base than an acid at that position. This is consistent with the anticipated behavior of any first deprotonation of a diprotic acid in which the conjugate base HA^- hydrolysis:



The second equivalence point pH of 10.52 is more basic since A^{2-} is a relatively strong base. These are alkaline pH values which are indicative that phenolphthalein (transition range pH 8.2-10.0) is the right indicator to use in the first equivalency point but not the second.

4. How do the two equivalence point volumes relate to acid concentration?

In the case of a pure diprotic acid in which the protons have the same stoichiometric factor, the second equivalence point volume must be precisely twice the first equivalence point volume ($V_2 = 2 V_1$). $V_1 = 12.64 \text{ mL}$ and V_2 of approximately 13.0 mL are used in this experiment. The minor divergence indicates that the acid is not necessarily diprotic or that there might be some systematic error in determining the second equivalence point. The close correspondence confirms that the acid is basically diprotic with two titratable protons.

5. Evaluate the precision and accuracy of the standardization.

The standardization of NaOH was very precise with a RSD of 0.35% which is significantly lower than the 2 percent limit of proper analytical work. The confidence interval of 95% of ± 0.0009 M reflects a relative uncertainty of $\pm 0.9\%$. Such high accuracy implies accuracy in pipetting, endpoint determination, and proper technique. Since there is no known reference value of the NaOH solution, one cannot determine absolute accuracy, but the low random error indicates that the systematic errors are also small. The similarity of three independent titrations substantiates the validity of the finding.

Conclusion

This two-part experiment was able to illustrate several fundamental principles of volumetric acid-base analysis. NaOH solution was standardized in part 1 using a primary standard KHP to give $[\text{NaOH}] = (0.1066 \pm 0.0009)$ M with great precision (RSD = 0.35%). The narrow confidence interval and replicate values supported appropriate volumetric procedure and correct endpoint detection through phenolphthalein indicator. The blank correction adequately reflected the effects of indicator consumption and dissolved CO_2 . Application of the standardized titrant to a diprotic acid of unknown identity in Part 2 gave a clear titration curve with two distinct equivalence points at 12.64 mL (pH 7.82) and 13.0mL (pH 10.52). The unknown acid concentration was calculated to be 0.0510 M based on the first volume of the equivalence point. Buffer region analysis produced estimated pKa values of 2.6 and 10.5, indicating that the unknown is probably a dicarboxylic acid with widely separated pKa values. Diprotic character was confirmed by the almost 2:1 ratio of the volumes of the equivalence points. Both experiments demonstrated the level of precision at which volumetric methods could be quantified and showed the significance of technique in volumetric methods.